

Exercise I.

[20 marks]

Compute the following integrals:

1. $\int \sqrt{1-x^2} dx,$

2. $\int x \frac{\sin x}{(\cos x)^3} dx,$

3. $\int \frac{x^4 - x^3}{x+1} dx,$

4. $\int \frac{\sqrt{x} - \sqrt[4]{x}}{\sqrt{x} + \sqrt[4]{x}} dx.$

Exercise II.

[20 marks]

Study the natures of the following series:

1. $\sum_{n=2}^{\infty} \frac{\ln n}{n^2}.$

2. $\sum_{n=2}^{\infty} (-1)^n \frac{n}{(\ln n)^5}.$

3. $\sum_{n=1}^{\infty} \frac{2^n}{3^n - 1},$

4. $\sum_{n=1}^{\infty} \left(\frac{n^2 + 1}{2n^2 + 3} \right)^n.$

Exercise III.

[20 marks]

Let $\alpha \in \mathbb{R}$. Study the natures of the following improper integrals:

1. $\int_1^{\infty} \frac{1}{x+x^\alpha} dx,$

2. $\int_0^1 \frac{1}{(\sin x)^\alpha \sqrt{|\ln x|}} dx,$

3. $\int_0^{\infty} \sin(2x) dx,$

4. $\int_0^{\infty} \frac{\cos x}{1 - \cos(\sqrt[4]{x}) + x^2} dx.$

Exercise IV.

[25 marks]

1. Let $f : \mathbb{R} \mapsto \mathbb{R}$ be a continuous function. Calculate the following limits:

a. $\lim_{n \rightarrow \infty} \left(\prod_{k=1}^n e^{\frac{n^2+k^2}{n^2}} \right)^{\frac{1}{n}}.$

b. $\lim_{x \rightarrow 0} \left(x \int_0^x \frac{1}{x^2 + t^2} f(t) dt \right).$

2. Let $g(x) = \int_{2\sqrt{x}}^{x^2} \ln(t-2) dt$. Compute $g'(x)$ and specify the domain of differentiability of g .

Exercise V.

[15 marks]

Let $p \in \mathbb{R}$ and $\Gamma(p) = \int_0^{\infty} x^{p-1} e^{-x} dx$.

1. Study the nature of $\Gamma(p)$.
2. Find a relation between $\Gamma(p+1)$ and $\Gamma(p)$.
3. Deduce that $\Gamma(n+1) = n!$ for all $n \in \mathbb{N}$.

Exercise I. Compute the following integrals:

$$1. \int \sqrt{1-x^2} dx = \int \sqrt{1-\sin^2 t} \cos t dt$$

$$= \int |\cos t| \cos t dt$$

$$= \int \cos^2 t dt$$

$$= \frac{1}{2} \int [1 + \cos 2t] dt = \frac{1}{2} t + \frac{1}{4} \sin 2t + C$$

$$= \frac{1}{2} t + \frac{1}{2} \sin t \cos t + C \stackrel{(1B)}{=} \frac{1}{2} \arcsin x + \frac{1}{2} x \sqrt{1-x^2} + C.$$

- $x = \sin t = u(t)$, $dx = \cos t dt$
- u diff and st $\uparrow$ on $J = [-\frac{\pi}{2}, \frac{\pi}{2}]$
- f cont on $I = [-1, 1]$
- $u(J) = [-1, 1] \subseteq I$

$$2. \int x \cdot \frac{\sin x}{(\cos x)^3} dx = \frac{x}{2 \cos^2 x} - \frac{1}{2} \int \frac{1}{\cos^2 x} dx$$

$$= \frac{x}{2 \cos^2 x} - \frac{1}{2} \tan x + C.$$

Int by parts:

$$u = x \quad v' = \frac{\sin x}{(\cos x)^3}$$

$$u' = 1 \quad v = \frac{1}{2 \cos^2 x}$$

$$3. \int \frac{x^4 - x^3}{x+1} dx = \int \left[x^3 - 2x^2 + 2x - 2 + \frac{2}{x+1} \right] dx$$

$$= \frac{x^4}{4} - \frac{2}{3} x^3 + x^2 - 2x + 2 \ln|x+1| + C$$

$$\begin{array}{r|l} x^4 - x^3 & x+1 \\ \hline -x^4 + x^3 & x^3 - 2x^2 + 2x - 2 \\ \hline -2x^3 & \\ +2x^3 + 2x^2 & \\ \hline 2x^2 & \\ -2x^2 + 2x & \\ \hline 2x & \\ +2x & \\ \hline 4 & \end{array}$$

$$4. \int \frac{\sqrt{x} - \sqrt[4]{x}}{\sqrt{x} + \sqrt[4]{x}} dx = \int \frac{t^2 - t}{t^2 + t} (4t^3) dt$$

$$x = t^4 \Rightarrow dx = 4t^3 dt$$

$$= 4 \int \frac{t^4 - t^3}{t+1} dt \stackrel{\text{using}}{=} t^4 - \frac{8}{3} t^3 + 4t^2 - 8t + 8 \ln|t+1| + C$$

$$\stackrel{t=x^{1/4}}{=} x - \frac{8}{3} x^{3/4} + 4x^{1/2} - 8x^{1/4} + 8 \ln|x^{1/4}+1| + C$$

Exercise II. Study the natures of the following series:

$$1. \sum_{n=2}^{\infty} \frac{\ln n}{n^2} = \sum_{n=2}^{\infty} \frac{1}{n^2 \ln n}; \quad \text{Conv, Bertrandian } \alpha=2>1.$$

$$2. \sum_{n=2}^{\infty} (-1)^n \frac{n}{(\ln n)^5}; \quad \lim_{n \rightarrow \infty} (-1)^n \frac{n}{(\ln n)^5} \neq 0 (\exists) \Rightarrow \text{The series diverges.}$$

$$3. \sum_{n=1}^{\infty} \frac{2^n}{3^n - 1} = \sum_{n=1}^{\infty} u_n; \quad \left| \frac{u_{n+1}}{u_n} \right| = \frac{2^{n+1}}{3^{n+1} - 1} \times \frac{3^n - 1}{2^n} \stackrel{(2)}{=} \frac{2(3^n - 1)}{3^{n+1} - 1} \stackrel{(1)}{\sim} \frac{2 \cdot 3^n}{3^{n+1}} \stackrel{(1)}{\rightarrow} \frac{2}{3} < 1$$

The series conv absol (d'Alembert Ratio test)

$$4. \sum_{n=1}^{\infty} \left(\frac{n^2+1}{2n^2+3} \right)^n = \sum_{n=1}^{\infty} v_n; \quad |v_n|^{1/n} = \frac{n^2+1}{2n^2+3} \stackrel{(3)}{\rightarrow} \frac{1}{2} < 1. \quad \text{The series conv absol.} \quad (2)$$

Exercise III. Let $\alpha \in \mathbb{R}$. Study the natures of the following improper integrals:

$$1. \int_1^{\infty} \frac{1}{x+x^\alpha} dx \quad \begin{matrix} \nearrow \text{near } \infty \\ 0 < \frac{1}{x+x^\alpha} \sim \frac{1}{x^\alpha} \end{matrix} \quad \begin{cases} \frac{1}{x^\alpha} & \text{if } \alpha > 1 \\ \frac{1}{2x} & \text{if } \alpha = 1 \\ \frac{1}{x} & \text{if } \alpha < 1 \end{cases} \quad \begin{matrix} \textcircled{1} \\ \textcircled{1} \\ \textcircled{1} \end{matrix}$$

By equivalent test; The imp int conv iff $\alpha > 1$. $\textcircled{2}$

$$2. I = \int_0^1 \frac{f(x)}{(\sin x)^\alpha \sqrt{|\ln x|}} dx = \int_0^{1/2} f + \int_{1/2}^1 f = I_1 + I_2$$

near 0^+ : $f(x) \sim \frac{1}{x^\alpha |\ln x|^{1/2}}$. I_1 conv iff $\alpha < 1$ (Bertrand) $\textcircled{1}$

near 1^- : $f(x) = \frac{1}{(\sin x)^\alpha (|\ln(1+x-1)|)^{1/2}} \sim \frac{1}{(\sin x)^\alpha (1-x)^{1/2}}$. I_2 conv (Riemann) $\textcircled{1}$

Thus I conv iff $\alpha < 1$. $\textcircled{1}$

$$3. \int_0^{\infty} \sin(2x) dx \stackrel{\textcircled{2}}{=} \lim_{x \rightarrow \infty} \int_0^x \sin(2x) dx \stackrel{\textcircled{2}}{=} \lim_{x \rightarrow \infty} \left[-\frac{1}{2} \cos 2x \right]_0^x = -\frac{1}{2} \lim_{x \rightarrow \infty} [\cos 2x - 1] \nexists$$

Hence the imp int diverges.

$$4. I = \int_0^{\infty} \frac{\cos x}{1 - \cos(\sqrt[4]{x}) + x^2} dx = \int_0^1 f + \int_1^{\infty} f = I_1 + I_2$$

near 0^+ : $f(x) \sim \frac{1}{1 - [1 - \frac{\sqrt{x}}{2}]} = 2 \cdot \frac{1}{x^{1/2}}$. Hence I_1 conv (Riemann) $\textcircled{1}$

near ∞ : $|f(x)| = \frac{|\cos x|}{1 - \cos \sqrt[4]{x} + x^2} \leq \frac{1}{x^2}$. Hence I_2 conv (compt + Riem) $\textcircled{1}$

Thus I is $\textcircled{1}$ convergent.

Exercise IV.

1. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a cont fcn. Calculate the following:

$$a. \lim_{n \rightarrow \infty} \left(\prod_{k=1}^n e^{\frac{n^2+k^2}{n^2}} \right)^{1/n} \stackrel{\textcircled{2}}{=} \lim_{n \rightarrow \infty} \left(e^{\sum_{k=1}^n \frac{n^2+k^2}{n^2}} \right)^{1/n} \stackrel{\textcircled{1}}{=} \lim_{n \rightarrow \infty} e^{\frac{1}{n} \sum_{k=1}^n 1 + (\frac{k}{n})^2} = e^{\int_0^1 (1+x^2) dx}$$

$$= e^{\left[x + \frac{x^3}{3} \right]_0^1} = e^{1 + \frac{1}{3}} = e^{4/3}$$

$f(x) = 1+x^2$ cont $\textcircled{1.5}$
hence Riem. integ
on $[0,1]$.

$$b. \lim_{x \rightarrow 0} \left(x \int_0^x \frac{1}{x^2+t^2} f(t) dt \right)$$

$$\stackrel{\text{ISCHURIT}}{=} \lim_{x \rightarrow 0} \left(x f(c_x) \int_0^x \frac{1}{x^2+t^2} dt \right)$$

$$= \lim_{x \rightarrow 0} \left(x f(c_x) \frac{1}{x} \arctan \frac{t}{x} \Big|_{t=0}^{t=x} \right)$$

$$= \lim_{x \rightarrow 0} f(c_x) [\arctan 1 - \arctan 0]$$

$$= f(0) \cdot \frac{\pi}{4} \quad (\because f \text{ cont at } 0).$$

- $\textcircled{1}$ • $f \in C([0,1])$ ($\because f \in C(\mathbb{R})$)
- $\textcircled{1}$ • $g \in R([0,1])$ ($\because g$ cont on $[0,1]$)
- $\textcircled{1.5}$ • $g > 0$ on $[0,1]$
- $\textcircled{1}$ • $0 \leq c_x \leq x$
- as $x \rightarrow 0$, $c_x \rightarrow 0$.

2. Let $g(x) = \int_{2\sqrt{x}}^{x^2} \ln(t-2) dt$. Compute $g'(x)$ and specify the domain of diff of g .

• $g'(x) \stackrel{(3)}{=} u'(x)f(u(x)) - v'(x)f(v(x)) \stackrel{(2)}{=} 2x \ln(x^2-2) - \frac{1}{\sqrt{x}} \ln(2\sqrt{x}-2).$

• f is cont on the interval $j =]2, +\infty[$. ($t-2 > 0 \Leftrightarrow t > 2 \Leftrightarrow t \in]2, +\infty[$).

• $u, v: I \rightarrow j$ diff:

$\uparrow u, v$ diff on $\mathbb{R} \cap]0, +\infty[=]0, +\infty[= I_1 \stackrel{(2)}{}$

$\hookrightarrow u(x), v(x) \in j \Leftrightarrow x^2 > 2 \text{ and } 2\sqrt{x} > 2 \Leftrightarrow x \in (]-\infty, -\sqrt{2}[\cup]\sqrt{2}, +\infty[) \cap]1, \infty[$

$\stackrel{(2)}{\Leftrightarrow} x \in]\sqrt{2}, +\infty[= I_2$

Hence $I = I_1 \cap I_2 \stackrel{(1)}{=}]\sqrt{2}, +\infty[$ ensures $u, v: I \rightarrow j$ diff.

So the domain of diff of g is $I =]\sqrt{2}, +\infty[$.

Exercise V. Let $p \in \mathbb{R}$ and $\Gamma(p) = \int_0^\infty x^{p-1} e^{-x} dx$.

1. Study the nature of $\Gamma(p)$.

near 0^+ : $f(x) = x^{p-1} e^{-x} \stackrel{(1)}{\underset{0^+}{\sim}} x^{p-1} = \frac{1}{x^{1-p}} > 0$. $\Gamma_1 = \int_0^1 f$ conv iff $1-p < 1$ iff $p > 0$. (1)

near ∞ : $x^2 f(x) \stackrel{(1)}{=} x^2 x^{p-1} e^{-x} \xrightarrow{x \rightarrow \infty} 0$ & $\alpha = 2 > 1 \xRightarrow{x^{\alpha} \text{ tel}} \Gamma_2 = \int_1^\infty f$ converges. (1)

$\Gamma(p)$ conv iff Γ_1 & Γ_2 conv iff $p > 0$. (1)

2. Find a relation between $\Gamma(p+1)$ and $\Gamma(p)$.

$\Gamma(p+1) \stackrel{(1)}{=} \int_0^\infty \underbrace{x^p}_{u'} \underbrace{e^{-x}}_{v'} dx \stackrel{(2)}{=} \underbrace{-x^p e^{-x}}_{\substack{\text{finite} \\ \text{at } 0}} \Big|_0^\infty - \int_0^\infty p x^{p-1} (-e^{-x}) dx \stackrel{(1)}{=} p \Gamma(p)$

3. Deduce that $\Gamma(n+1) = n!$ for all $n \in \mathbb{N}$.

By induction:

• $\Gamma(1) \stackrel{(2)}{=} \int_0^\infty e^{-x} dx = 1 = 0!$

• $\Gamma(n) = (n-1)! \Rightarrow \Gamma(n+1) \stackrel{(2)}{\rightarrow} p \Gamma(n) \stackrel{(1)}{=} n!$

$\Gamma(n+1) \stackrel{(2)}{=} n \Gamma(n)$
 $= n(n-1) \Gamma(n-1)$
 $= n(n-1)(n-2) \dots \Gamma(1)$
 $= n(n-1)(n-2) \dots 1 = n!$